Machine Learning for Spectral Analysis

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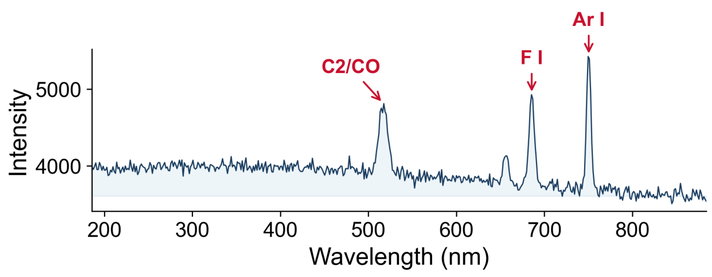
1. Introduction

"How can we automate multi-species identification and plasma state classification from high-dimensional OES data?"

Core contributions:

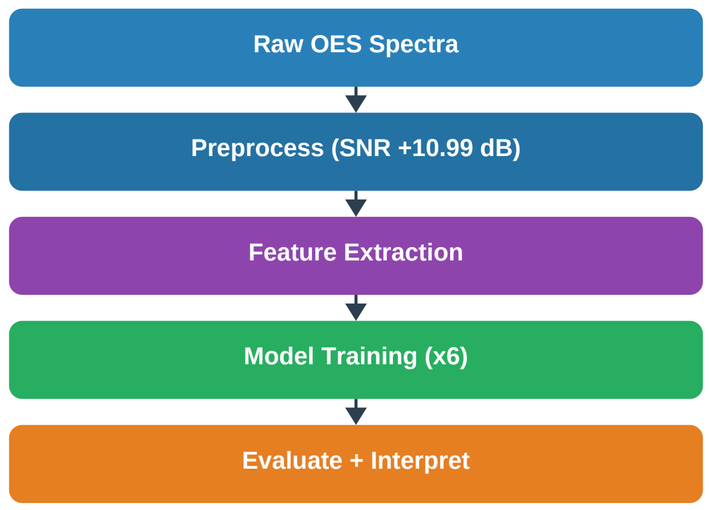
- **Species identification** via NMF + NIST matching
- **Plasma state classification** (6 models compared)

Also: actinometry, Boltzmann T_exc, temporal DTW clustering



Dataset	Channels	Task
BOSCH RIE	3,648	Plasma ON/OFF
Mesbah CAP	51	T_rot / T_vib

2. Methodology



Key Decisions:

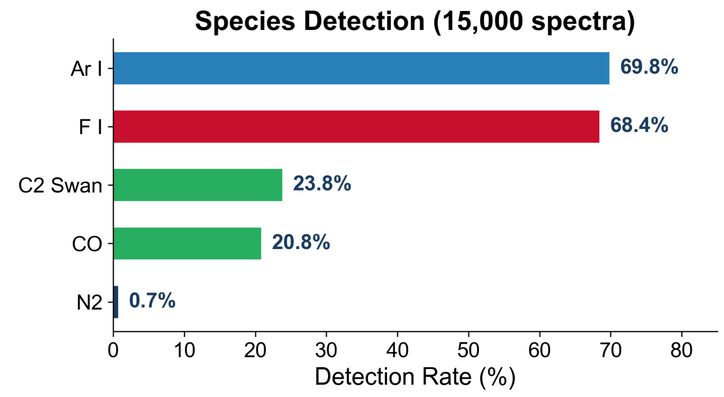
→ GroupKFold CV prevents sample leakage

→ Balanced class weights (12.1% OFF minority)

→ NMF enforces non-negative physics constraints

3. Species Identification

NMF (X = W . H) decomposes mixed spectra into pure-species components.

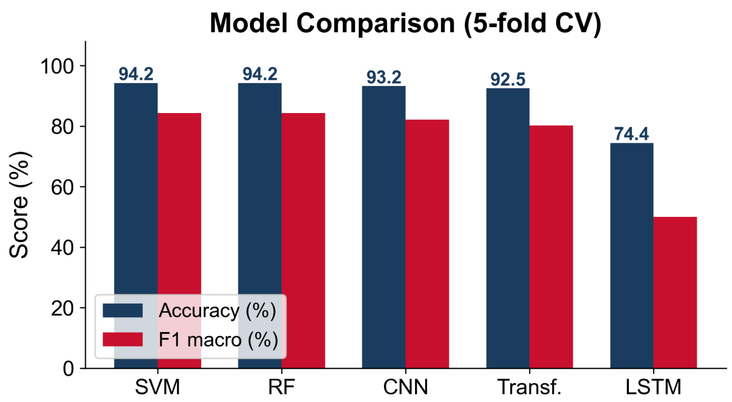


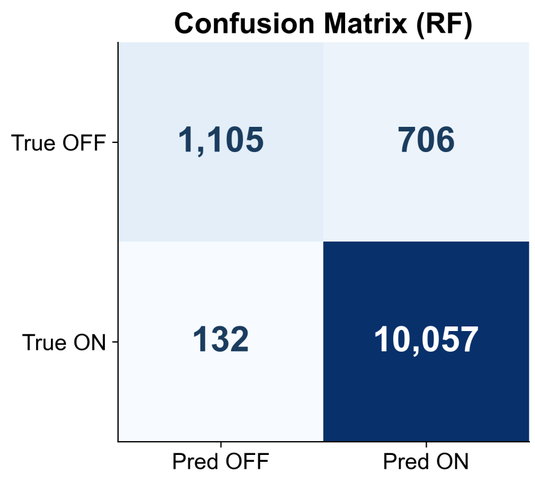
Species	Lines (nm)	Origin
F I	685.6, 703.7	SF6 etchant
Ar I	696.5, 750.4	Carrier gas
C2 Swan	516.5, 563.6	C4F8 indicator

Validation: NMF peaks at 684.4 nm (=F I) and 515.1 nm (=C2) confirm NIST database independently.

4. Classification Results

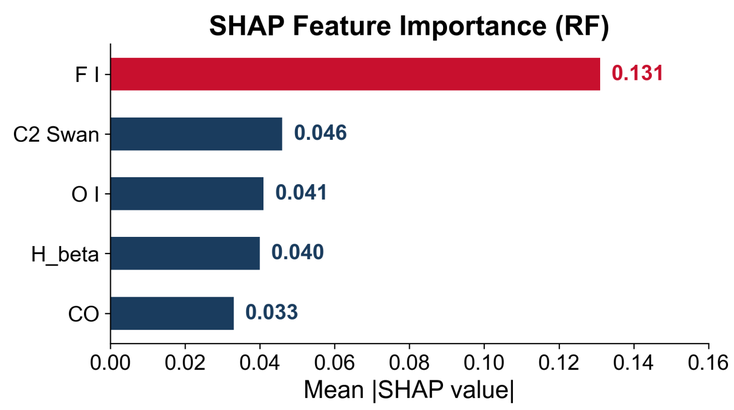
F1 macro = 0.843 | Accuracy = 94.2%



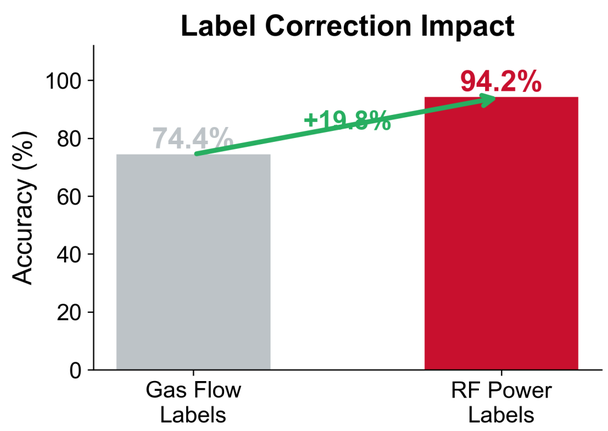


Note: OFF recall=61% reflects 12.1% class imbalance. Balanced weights applied but minority detection remains challenging.

5. Interpretability & Physics



SHAP confirms F I (fluorine radical) as the primary discriminator — consistent with SF6 dissociation chemistry.



Key discovery: Initial gas-flow labels produced only 74% accuracy. Per-species intensity analysis revealed etch/passivation spectral difference < 0.35 std — indistinguishable. Switching to RF-power labels (plasma ON/OFF) improved to **94%**, proving **data quality, not model complexity, was the real bottleneck**.

Boltzmann T_exc = 13,334 K (6 Ar I lines, 696-772 nm).

6. Conclusions

Key Takeaways:

✓ Automated pipeline identifies **13 plasma species** from 3,648-channel OES and classifies plasma state at **F1 = 0.843**.

✓ **Data quality trumps model complexity:** label correction (gas-flow → RF-power) yielded +19.8% accuracy gain — larger than any model architecture change.

✓ SHAP analysis validates that ML predictions align with **plasma physics** (F I = primary etchant radical).

Future Work: Extend to multi-class plasma state tracking; implement real-time streaming inference for industrial deployment.

References:

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[8] BOSCH dataset: Zenodo #17122442